Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

This tells that R2 is only sending Hello packets out of S0/0/0 (towards R1). Interface S0/0/1 is still a passive interface, so it suppresses Hello packets, preventing R2 from becoming an OSPF neighbor with R3.

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

The cost is 65. It is calculated by adding the cost of R3's S0/0/1 interface (64) and the cost of R2's G0/0 interface (1).

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

To allow OSPF to accurately calculate the cost for high-speed links. By default, OSPF uses 100 Mbps as the reference bandwidth, meaning any link 100 Mbps or faster (like Gigabit Ethernet) will have the same cost of 1.

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Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

Cost to 192.168.3.0/24 is 782: Derived from R1's S0/0/1 cost (781) plus R3's G0/0 cost (1).

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Cost to 192.168.23.0/30 is 845: Derived from R1's S0/0/1 cost (781) plus R3's S0/0/1 cost (64).

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

2

1562. With all serial interfaces set to 128 Kbps, each interface cost becomes 781. The path crosses two serial links (R1's and the neighbor's), totaling 781 + 781 = 1562.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF chooses the route with the lowest cost. The path via R2 has a total cost of 1563 (781 + 781 + 1). This is lower than the direct path via R3, which has a total cost of 1566 (1565 + 1).

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Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   Controlling the Router ID ensures the stability of the OSPF routing domain. It allows network administrators to easily identify specific routers within routing tables and neighbor relationships. Furthermore, in multi-access networks, the Router ID is the deciding factor in the DR/BDR (Designated Router/Backup Designated Router) election process.

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2. Why is the DR/BDR election process not a concern in this lab?

   The routers in this lab are connected via point-to-point serial links, a network type that does not require DR/BDR elections. Additionally, the Gigabit Ethernet interfaces are connected directly to end devices (PCs) rather than other OSPF routers.

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3. Why would you want to set an OSPF interface to passive?

   Setting an interface connected to a LAN or end devices to passive prevents OSPF routing updates from being unnecessarily broadcasted onto that network. This not only saves network bandwidth and router CPU resources but also enhances security by preventing internal routing information from being exposed to unauthorized devices.

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4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   The use of OSPF Areas (such as the backbone Area 0 and other regular areas) creates a hierarchical structure. This design effectively reduces routing table sizes and limits the flooding scope of Link-State Advertisements (LSAs), thereby building highly scalable networks.

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5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   network 10.0.0.0 0.255.255.255 area 0

   4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   The EIGRP route will be used. When comparing different routing protocols, a router prioritizes the Administrative Distance (AD) over the metric. EIGRP has an AD of 90, which is lower (more trustworthy) than OSPF (110) and RIP (120). Therefore, the router will select the EIGRP path.

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